
EMBEDDED HACKING

Reverse Engineering the Raspberry Pi Pico 2

OPERATION IRON VEIN - Requirements & Grading Criteria

ARM Cortex-M33 • GDB • Ghidra

September 21, 2026

For educational and defensive security research only

OPERATION IRON VEIN

REQUIREMENTS & GRADING CRITERIA

TARGET: NorthPharma SCADA pipeline valve controller

ARTIFACT: ACT-III.bin / ACT-III.uf2 (compromised)

CREW: FROSTLINE

OPERATIVE: NIGHTINGALE

Project Overview

NorthPharma runs the cold chain that keeps a vaccine lot viable, and its SCADA pipeline valve controller built on a Pico 2 is the hand at the end of the line. A contractor called **FROSTLINE** planted an implant in the controller image: a covert beacon, a logic bomb, a reserved-sector persistence marker, and an inverted valve command authorization verdict. Operative **NIGHTINGALE** recovered the compromised image as `ACT-III.bin`.

Students are the reverse-engineering reserve. They reverse engineer `ACT-III.bin` with Ghidra, find and patch all four defects, defeat the CoreDebug `DHCSR` anti-debug under GDB to observe the payload, export a corrected image, flash it to a real Pico 2, and prove the corrected behavior on the breadboard. The machine check is `scripts/verify_ctf.py`.

The challenge is a standalone capstone exercise and contains no answer, constant, address, bug, or patch belonging to any other course assignment.

Learning Objectives

- Decode an ARM Cortex-M33 vector and boot table and identify the reset handler and initial stack pointer.
- Map a stripped firmware image into modules by tracing calls from `main` and the monitor loop.
- Locate four corrupted bytes: a covert beacon gate, a logic-bomb arming gate, a persistence gate, and an authorization verdict branch.
- Analyze `cbz`, `cbnz`, `beq`, and `bne` condition semantics and branch inversion.
- Explain why a periodic magic preamble is a detection opportunity and why a trigger can detonate through the actuator path.
- Read CoreDebug `DHCSR`, explain the anti-debug trap, and defeat it under GDB.
- Explain why authentication is not authorization and why a verdict must be verified before the actuator moves.

Students must use only the course concepts: ARM registers, stack behavior, USB-CDC and UART consoles, GDB, Ghidra static analysis and binary patching, vector tables, reset startup, XIP, Thumb addressing, condition-code analysis, stateful security, and the Argon2id plus XChaCha20-Poly1305 authenticated envelope.

Deliverables Checklist

#	Deliverable	Format	Criterion
1	Ghidra project screenshot	PNG/JPG	Task 1
2	Vector table and boot table	Inside ACT-III-Answers.md	Task 1
3	main and monitor-loop table	Inside ACT-III-Answers.md	Task 1
4	Module map	Inside ACT-III-Answers.md	Task 1
5	Beacon evidence and patch	Inside ACT-III-Answers.md	Task 2
6	Logic bomb evidence and patch	Inside ACT-III-Answers.md	Task 3
7	Anti-debug GDB proof, persistence evidence, and patch	Inside ACT-III-Answers.md	Task 4
8	Authorization evidence and patch	Inside ACT-III-Answers.md	Task 5
9	ACT-III_fixed.bin	BIN file	Task 6
10	ACT-III_fixed.uf2	UF2 file	Task 6
11	Hardware proof and reflection	Inside ACT-III-Answers.md	Task 6

Required Tools and Equipment

Tool	Purpose
Raspberry Pi Pico 2	Isolated target node
Debug Probe (OpenOCD)	SWD connection for GDB inspection and the anti-debug work
arm-none-eabi-gdb	Runtime breakpoints, DHCSR clearing, and payload observation
Ghidra	Static analysis and binary patching
Python 3 with uf2conv.py	UF2 conversion and artifact checks
DHT11, 1602 I2C LCD, RYL998, IR receiver, SG90 servo, 3 LEDs, ESTOP button	Breadboard hardware proof
ACT-III.bin and ACT-III.uf2	Supplied compromised artifacts

Console settings: **USB-CDC virtual COM port, 115200 baud, 8 data bits, no parity, 1 stop bit.** Radio UART settings: **UART1, 115200, network ID 18.**

Artifact Identity

The instructor-issued artifact hashes are:

ACT-III.bin	a0f9206fdc4a33ecd24db15d179c31c40ec8abbc358489fe61a7d45dcf2d2b48
ACT-III.uf2	aed2e62f127f0ceb9d60621ef02c16850669679d1939f9ec37269cb7bd18aff4
ACT-III_fixed.bin	bac3e278ce9ba9d0c166065b0218e71f3e3eb2ff0f80fee5f439e93ab058fa0c
ACT-III_fixed.uf2	fb53ca66ec00c448bb28f14a8bc9d9bb71161f8b448c8b8c13cd8a9b00ea7653

The verifier checks the `ACT-III.bin` and `ACT-III_fixed.bin` hashes specifically, asserts the four fixed bytes, and requires that only those four offsets differ between the two `.bin` images. Both `.bin` images are 50,732 bytes and both `.uf2` images are 102,400 bytes.

Grading Rubric - Detailed Breakdown

Task 1: Setup and Initial Analysis (10 points)

Criterion	Points	Full credit	Partial credit	No credit
[DOCUMENT] Ghidra project created with the correct name and settings	2	Project <code>IronVein_Investigation</code> , raw binary import	One item off	Not set up
[DOCUMENT] Processor configured as ARM Cortex 32 little endian default	2	Screenshot shows the correct processor	Wrong language	Missing
[DOCUMENT] Base address set to 0x10000000	2	Base <code>0x10000000</code>	Wrong base	Missing
[DOCUMENT] Vector table, initial stack pointer, and reset handler identified	2	Base <code>0x10000000</code> , initial SP <code>0x20082000</code> , reset handler <code>0x1000015D</code>	One missing	Not found
[DOCUMENT] main and the SCADA controller state machine (<code>monitor_step</code>) addresses identified	1	<code>main</code> <code>0x10000234</code> , <code>monitor_step</code> <code>0x1000659C</code>	One correct	Neither
[DOCUMENT] Module map identifies the valve, control, <code>valve_auth</code> , <code>implant</code> , and <code>monitor</code> anchors	1	At least one correct anchor per module	Partial	Missing

Task 2: Bug #1 The Beacon (20 points)

Criterion	Points	Full credit	Partial credit	No credit
[DOCUMENT] Located the beacon gate branch at 0x1000A245	5	Address and function identified	Approximate	Not found
[DOCUMENT] Documented <code>cbz</code> versus <code>cbnz</code> and the every-8-ticks DE AD BE EF beacon	5	Correct branch semantics and beacon detail	Partial	Wrong
[DOCUMENT & PATCH] Patched 0xB9 to 0xB1 so the beacon gate is closed	7	Byte <code>0xB9</code> changed to <code>0xB1</code>	Wrong byte	Not patched
[DOCUMENT] Explained how to prove the covert beacon stopped	3	Periodic magic frame no longer observed	Vague	Missing

Task 3: Bug #2 The Logic Bomb (20 points)

Criterion	Points	Full credit	Partial credit	No credit
[DOCUMENT] Located the arming gate branch at 0x1000A30B	5	Address and function identified	Approximate	Not found
[DOCUMENT] Documented the FROSTLNE arming magic and the 3-tick detonation	5	Correct magic, delay, and detonation path	Partial	Wrong
[DOCUMENT & PATCH] Patched 0xB9 to 0xB1 so the logic bomb is disarmed	7	Byte 0xB9 changed to 0xB1	Wrong byte	Not patched
[DOCUMENT] Explained that the bomb closes the valve through the actuator path	3	Direct actuator call, no authorization	Vague	Missing

Task 4: Bug #3 The Persistence Marker (20 points)

Criterion	Points	Full credit	Partial credit	No credit
[DOCUMENT] Located the persistence gate branch at 0x1000A29F	5	Address and inlined persist path identified	Approximate	Not found
[DOCUMENT] Documented the CoreDebug DHCSR anti-debug and how it is defeated under GDB	5	0xE000EDF0, C_DEBUGEN and C_HALT, and a real defeat method	Partial	Wrong
[DOCUMENT & PATCH] Patched 0xD1 to 0xD0 so no marker is written to 0x103FF000	7	Byte 0xD1 changed to 0xD0	Wrong byte	Not patched
[DOCUMENT] Explained the write-once marker byte 0xC7 and the Act IV persistence preview	3	Marker, reserved sector, write-once latch, persistence lesson	Vague	Missing

Task 5: Bug #4 The Valve Command Authorization (20 points)

Criterion	Points	Full credit	Partial credit	No credit
[DOCUMENT] Located the authorization branch at 0x100074E9	5	Address and function identified	Approximate	Not found
[DOCUMENT] Documented the authorization verdict inversion and the branch condition	5	Reject when the verdict is false	Partial	Wrong
[DOCUMENT & PATCH] Patched 0xD1 to 0xD0 so unauthorized and replayed commands are rejected	7	Byte 0xD1 changed to 0xD0	Wrong byte	Not patched
[DOCUMENT] Explained that unauthenticated and replayed valve commands must be rejected	3	The actuator must see only an authorized verdict	Vague	Missing

Task 6: Export and Verify (10 points)

Criterion	Points	Full credit	Partial credit	No credit
[PATCH] Exported ACT-III_fixed.bin from Ghidra	2	Valid patched binary	Corrupt	Not submitted
[PATCH] Converted to ACT-III_fixed.uf2 with the correct base and family	2	<code>--base 0x10000000 --family 0xe48bff59</code>	Wrong flags	Not submitted
[DOCUMENT] scripts/verify_ctf.py passes and hardware proves the correct behavior	3	Verifier passes and the hardware proof is shown	Partial proof	No proof
[DOCUMENT] Reflection maps each of the four defects to a real-world control-system failure	3	Specific mapping for all four	Partial	Missing

Common Pitfalls

Pitfall	Consequence	Avoidance
Reading the beacon gate backwards	The covert frame keeps transmitting	Accept only on the neutralization branch (<code>cbz</code> , <code>0xB1</code>)
Confusing <code>cbz</code> and <code>cbnz</code> at <code>0xA30B</code>	The <code>FROSTLINE</code> magic still arms the bomb	Neutralize only when the gate is clear (<code>cbz</code> , <code>0xB1</code>)
Missing that the persist path is inlined	Cannot find the marker gate	Look inside <code>implant_tick</code> at <code>0x1000A29F</code>
Patching the shipped image before observing the payload	You never prove the marker write	Defeat <code>DHCSR</code> under GDB first, then patch the artifact
Fabricating the GDB session	Verification fails	Show the command sequence and the real observed code path
Treating the anti-debug as a defect to patch	Wasted effort; it is identical in both images	Defeat it in a scratch copy or with GDB, then patch the real defect
Missing that the authorization branch is a verdict	Unauthenticated commands still reach the valve	Accept only when the verdict is true (<code>beq</code> to reject, <code>0xD0</code>)
Forgetting UF2 conversion	Raw binary will not flash	Use <code>uf2conv.py</code> with family <code>0xe48bff59</code>

How To Breadboard

Device	Pin on device	Pico 2 GPIO	Notes
DHT11 process sensor	DATA	GP4	10 kOhm pull-up to 3.3 V if the module needs it
1602 LCD	SDA	GP2	I2C1, backpack address <code>0x27</code>
1602 LCD	SCL	GP3	I2C1, 100 kHz

Device	Pin on device	Pico 2 GPIO	Notes
1602 LCD	VCC / GND	VBUS 5 V / GND	The backpack needs 5 V, not 3.3 V
RYLR998	RX	GP8 (Pico TX)	UART1, 115200, network ID 18
RYLR998	TX	GP9 (Pico RX)	UART1
IR receiver	OUT	GP5	VS1838B, internal pull-up enabled
Servo	signal	GP14	PWM 50 Hz; 1000 uF bulk cap across servo 5 V and GND
Red LED	anode	GP16	VALVE FAULT, 220 to 330 ohm to GND
Yellow LED	anode	GP17	COMMAND PENDING, 220 to 330 ohm to GND
Green LED	anode	GP18	VALVE NOMINAL, 220 to 330 ohm to GND
Emergency-stop button	leg 1	GP15	Internal pull-up; leg 2 to GND, never to 3.3 V
Onboard LED	built in	GP25	Heartbeat
Debug Probe	SWCLK / SWDIO / GND	debug header	For GDB only

Use 3.3 V logic on every GPIO. The only 5 V connection is the LCD backpack supply. Keep the 1000 uF capacitor on the servo rail.

Memory Map Reference

Region	Address	Purpose
Bootrom	0x00000000	Immutable boot code
Flash/XIP	0x10000000	Vector table, code, rodata, data image
SRAM	0x20000000	Stack and writable state
CoreDebug_DHCSR	0xE000EDF0	Anti-debug register read by the implant
Implant reserved sector	0x103FF000	Persistence marker target (sector)
Implant tick counter	0x200136F0	Incremented once per <code>implant_tick</code>
Implant arming flag	0x20013CF0	Set when <code>FR0STLNE</code> arms the bomb
Implant beacon gate	0x20013CF1	Gates the covert beacon frame
Implant trigger tick	0x200136F4	Tick at which an armed bomb detonates
Implant command gate	0x20013CEF	Gates <code>FR0STLNE</code> handling
Implant persistence gate	0x20013CF2	Gates the marker write
Implant marker latch	0x20013CF3	Write-once latch for 0x103FF000

The VA of any file offset is the file offset plus `0x10000000`.

Deadline & Submission

- Create a folder containing the Ghidra screenshot, `ACT-III_fixed.bin`, and `ACT-III_fixed.uf2`.
 - Write all written answers in `ACT-III-Answers.md` inside that folder.
 - Include the output of `python scripts/verify_ctf.py`.
 - ZIP the folder as `lastname-firstname-ACT-III.zip`.
 - Submit the ZIP before the posted deadline; late submissions lose 10 percent per day.
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Grade Scale

Grade	Percentage	Points
A+	97-100%	97-100
A	93-96%	93-96
A-	90-92%	90-92
B+	87-89%	87-89
B	84-86%	84-86
B-	80-83%	80-83
C	70-79%	70-79
F	0-69%	0-69

Academic Integrity

Use only the supplied Pico 2 and firmware. Do not connect the exercise to an operational pipeline control system, a pharmaceutical network, a public network, a military system, or a third-party device. This is a controlled, isolated educational exercise. All analysis and patches must be your own work; sharing binaries, addresses, keys, passphrases, or answers is a violation of the academic integrity policy.

Reference Material

Topic	Reference
ARM Cortex-M33 registers and stack	Course block 1
USB-CDC and UART console capture	Course block 2
Vector tables, reset startup, and XIP	Course block 3
Ghidra static analysis and binary patching	Course block 4

Topic	Reference
Covert channels and control flow	Course block 5
CoreDebug DHCSR and anti-debug	Course block 6
Reserved-flash persistence	Course block 7
Argon2id and XChaCha20-Poly1305 authenticated envelope	Course block 8
Authentication versus authorization and fail-safe policy	Course block 9